

2 Description

The STM32C55xxx devices are general purpose microcontrollers family (STM32C5 Series) based on the high-performance Arm® Cortex®-M33 32-bit RISC core. They operate at a frequency of up to 144 MHz.

The Cortex®-M33 core features a single-precision floating-point unit (FPU), that supports all the Arm® single-precision data-processing instructions and all the data types.

The Cortex®-M33 core also implements a full set of digital signal processing (DSP) instructions and a memory protection unit (MPU) that enhances the application security.

The devices embed high-speed memories (512-Kbyte flash memory and 128-Kbyte SRAM), and an extensive range of enhanced I/Os, peripherals connected to three APB buses, three AHB buses, and a 32-bit multi-AHB bus matrix.

The devices feature several protection mechanisms for embedded flash memory and SRAM: readout protection, write protection, and hide protection areas.

The devices embed several peripherals reinforcing security:

- HASH hardware accelerator
- True random number generator

The devices offer two 12-bit ADCs, one DAC channel, one comparator, a low-power RTC, two 32-bit general-purpose timers, two 16-bit PWM timers dedicated to motor control, four 16-bit general-purpose timers, two 16-bit basic timers, and one 16-bit low-power timer.

The devices also feature standard and advanced communication interfaces such as:

- Two I²Cs
- One I³C shared with I²C
- Three SPIs with multiplexed full-duplex I2S
- Three USARTs, two UARTs, and one low-power UART
- One FDCAN (available only in STM32C552x devices)
- One USB full-speed

The devices operate in the –40 to +125 °C (+140 °C junction) temperature ranges from a 2.7 to 3.6 V power supply.

A comprehensive set of power-saving modes allows the design of low-power applications.

The devices offer multiple packages from 32 to 100 pins.

See [Table 1](#) for the list of peripherals available for each part number.

Table 1. Device features and peripheral counts

Peripherals		STM32C55xKxT	STM32C55xKxU	STM32C55xCxT	STM32C55xCxU	STM32C55xRxT	STM32C55xRxTxJ	STM32C55xMxT	STM32C55xVxT
Flash memory (Kbytes)		512/256							
SRAM (Kbytes)		128 (including 64 with ECC)							
ICACHE (Kbytes)		8							
Flash memory area for EEPROM Emulation (Kbytes)		64							
One-time-programmable (Kbytes)		4.5							
Timers	General purpose	6							
	Advanced-control	2							
	Basic	2							
	Low-power	1							